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A Neuro-Symbolic Curriculum Framework for Contextualized Augmented Reality Interventions in Autism Behavioral and Cognitive Development

Abstract-Augmented Reality (AR) technologies, combined with Artificial Intelligence, are increasingly explored to enhance cognitive, behavioral, and social skill development in children with Autism Spectrum Disorder (ASD). However, current systems predominantly rely on black-box personalization models, lack transparent intervention planning, and demonstrate limited ecological transfer to real-world contexts. This research proposes NSCCA-AR (Neural-Symbolic Curriculum for Contextualized AR), a novel hybrid framework that integrates lightweight neural perception modules with a symbolic curriculum planner to deliver explainable, constraint-aware, and context-adaptive intervention sequences. The system employs Bayesian belief updating to track individualized skill progression across domains such as joint attention, emotion recognition, and turn-taking, while a symbolic task-graph planner generates safe and developmentally appropriate sequences under attention, fatigue, and safety constraints. AR overlays dynamically scaffold engagement through on-device inference, ensuring privacy preservation via ephemeral feature representations without storing raw sensory data. In preliminary simulations with clinically validated task models, NSCCA-AR demonstrated a 31.6% improvement in targeted skill acquisition, a 24.8% enhancement in engagement duration, and a 19.3% increase in ecological generalization compared to baseline AR interventions. Additionally, clinician trust scores improved by 27.1% due to the transparency afforded by explainable decision traces. This work establishes a foundational paradigm for shifting from isolated AR tasks to structured, goal-driven developmental curricula, while ensuring privacy, interpretability, and real-world transferability. Future clinical pilots will further evaluate longitudinal outcomes, caregiver acceptance, and adaptability across diverse severity levels of ASD.

Keywords-Autism Spectrum Disorder, Augmented Reality, Neuro-Symbolic AI, Curriculum Planning, Bayesian Skill Modeling, Cognitive Development, Behavioral Intervention, Explainable AI, Privacy-Preserving Sensing, Real-World Generalization

I Introduction

Individuals with Autism Spectrum Disorder (ASD) face a wide range of problems in social communication, emotion recognition and executive functioning. Generalization of treatment gains Children with ASD frequently have difficulties in generalizing acquired skills from therapeutic or experimental settings to everyday environments and this affects long-term effectiveness of interventions [1][2]. Conventional behavioural and cognitive treatments rely on controlled environments, expert guidance, and structured exercises with translation to naturalistic settings limited by resource requirements and redundancy for the caregiver [3][4]. Augmented Reality (AR) has clearly shown promise as a path for scaffolding learning in ecologically viable settings through the use of overlays of interactive cues on top of real world scenes [5][6]. However, while promising, the majority of current AR approaches

leverage opaque neural personalization models or heuristic parameter tuning that compromise interpretability, transparency and control over adaptive safety oversight [7][8]. Filling these gaps requires a hybrid model that combines the flexible, perceptual abilities of neural architectures with the structured, constraint enforcing reasoning capabilities of symbolic planning for facilitating developmental interventions that are transparent and contextually updateable. Existing AR-based ASD systems are seen to suffer from severe limitations in terms of flexibility, explainability and generalizability. Although deep neural models are excellent at recognizing multimodal cues, their gaze gesture facial recognition interpretable decisions and clinician interpretability is also impeded by their black-box nature [9][10]. On the other hand, symbolic planners give computationally interpretable deduction/justification and hierarchical task planning, but suffer from limited robust open-loop replanning for noisy or even unpredictable sensory environments [11][12]. Several other models[13][14] also assume fixed task hierarchies and do not account for temporal and the physiological state including level of attention variation, fatigue, cognitive overload etc. In addition, such systems do not update their knowledge of a child's progress in real time and as the level of skills evolves, then one has no way to recalibrate interventions [15][16]. As a consequence, the static and fragmented intervention models which emerge can lead to under workload for both dissatisfaction and combined with or 'over' training where ecological validity is diminished. To bridge this gap, we develop an intelligent curriculum model that dynamically adjusts to the learner's behavioral and cognitive trajectory while ensuring interpretability and safety through symbolic reasoning.

The proposed framework, Neuro-Symbolic Curriculum for Contextualized Augmented Reality (NSCCA-AR), establishes a unified structure which incorporates neural perception, probabilistic reasoning and symbolic task planning [17]. Such neural modules transform multimodal input streams, comprising visual, gestural and vocal signals, into representations of task-relevant states such as gaze direction, emotional state or turn-taking. Finally, a symbolic curriculum planner builds adaptive task sequences according to explicit development, attentional and safety related constraints. Bayesian belief updating keeps a probability distribution over the learner's ability levels in different domains—joint attention, emotion recognition, and reciprocal interaction—and enables real-time adjustment of task difficulty. As skill estimates change, the planner promotes or demotes tasks to ensure engagement with reduced frustration. All the inference is conducted on-device, and we keep only temporary feature embeddings to guarantee privacy preservation [18]. This blend of neuroplastic adaptability and symbolic transparency paves the way for transparent, context-sensitive, ethically sound developmental interventions in AR. Real-world deployment of this technique requires methods that are interpretable, scalable and capable to handle non-stationary learning environments. AR interventions need to overcome the constraints of static task design and demonstrate real generalization to daily behaviors [19][20]. The main issues are the coordination between perceptual inference and constraint-based reasoning, the reduction of sensory noise, overcoming conflicting adaptation signals and maintaining belief stability along non-stationary behavioural trajectories. The present proposal aims to overcome these challenges by developing an intelligent, privacy-preserving curriculum engine to support longitudinal developmental progress in children with ASD. Combining

cognitive modelling, symbolic logic, and neural perception processes the study advances a flexible, transparent and ethically-informed approach to technology-mediated developmental learning with practical implications for clinical practice and inclusive education. The main contributions of the paper are:

- A neuro-symbolic curriculum architecture combining neural perception, Bayesian belief tracking and symbolic planning for personalized and explainable AR-based ASD interventions.
- Privacy-preserving Inference Pipeline Design: We develop a privacy-preserving inference pipeline leveraging temporary feature representations to sustain contextual personalization without retaining raw sensory data.
- Experimental results on simulated clinical task models indicate that skill acquisition, engagement and ecological generalization were significantly enhanced when compared to the AR framework.

The rest of the paper is formatted as follows. Section 2 introduces related works in AR-based intervention for ASD and points out limitations of adaptability and opacity. The proposed NSCCA-AR framework is presented in Section 3, which includes neural perception, Bayesian belief tracking and symbolic curriculum planning. In Section 4, we describe the experimental setting with regard to simulation parameters, evaluation measures and baselines. Experimental study results are presented and analyzed in Section 5, concentrating on skill learning improvement; engagement with the player as well as the interpretability for those skills. The final section 6 concludes by summarizing the main results and considering future clinical and developmental extensions.

II Literature Review

Nekar et al. (2022) presented evidence of the benefits of AR-based cognitive–motor training to enhance executive function and reaction time in children with ASD. Their search emphasized motivational AR environments to enhance compliance and performance, however with few effects for restricted repetitive behaviors. This illustrates the promise of AR as a behavioral approach to personalizing targeting in an individualized intervention for domains of executive control [1]. Based on this concept of peace, Doulah et al. (2023) conducted a systematic review which showed an effect of AR interventions on social interaction and learning situation in ASD individuals. Their findings reinforced the adapted use of AR to develop communication, regulatory and daily living activities. The same review also highlighted methodological issues in prior studies, emphasizing the need for rigorous experimental control and more informative feedback [2]. Yakubova et al. (2023) carried out a scoping review on AR and its role in academic learning for students diagnosed with ASD and ID. The positive was that they did find evidence for literacy, maths and science learning; but the conclusion was that these did not rise to the level of an ‘established programme’ in education. This point was a good representation of the required validation before integrating AR as clinical teaching published in academic curriculum [3]. Koumpouros (2025) accounted for ten years of the development of AR applications within developmental domains. In all papers, the results indicated that AR mobile or wearable platforms significantly improved

social communication and academic outcomes. It should also be noted in the author's previous work that there still remained challenges of intervention for ER and AR which were hotly sold to be combined by AI to customize interventions [4]. Leveraging on these findings, Abdullah et al. (2024) investigated AR supported rehabilitation for autistic children in attention/object ranking domain. Their work showed that AR simulations of activities in the real world yielded more cognitive engagement and understanding, particularly for mild to moderate ASD. Interactive visual learning environments proved to be able to increase attention and performance in a therapeutic context [5].

Lee and Huang (2025) employed a mixed technique to incorporate concept mapping into AR/MR in autistic children for social reasoning. By creating a structured teaching model, they transformed fuzzy social concepts into visible, game-like interactions where students can practice better emotional comprehension and situational judgment. This highlighted a requirement to "strengthen or embed structured cognitive scaffolding within such tools" [6]. Focusing on the fundamentals of social knowledge, Yazdanian, Mote & Wigand (2025) only targeted AR and VR interventions for joint attention – a crucial developmental capacity in autism. Their systematic review confirmed that AR/VR training was more effective in improving joint attention, but they could not pool the data due to heterogeneity issues among outcome measures. It has been recommended that a standardised outcome measure is required in order to progress future research of social-communication outcomes [7]. Consistent with these findings, Lledó and colleagues [30] have previously found that Delta-9-THC increased D2 like receptor mRNA expression. (2022) has investigated the use of AR in teaching handwashing and other ADLs for children with ASD. Through their quasi-experimental study, they demonstrated that participants showed more eye contact and motivation after the AR-based training. AR tailored to the learner has been proposed as a solution that can assist transfer of cognitive and behavioural learning gains [8]. Similarly, Pérez-Fuster et al. (2022) utilized AR through a Pictogram Room intervention to support JA responding in children with ASD. Using a recent single-case experimental design, they demonstrated that these effects were durable and even transferred to a new context. This was an initial indication that AR induced changes in fundamental social cognition [9]. Zhao et al. (2024) which compared XR applications and models in children with ASD through communication development and self-regulation. The findings confirmed that adaptive learning with multimodal interaction and AI-based personalization may be possible with XR. The development effects must be homogenous to ensure that ROS care MR environmentally does not somehow reduce efficiency [10] but house the writers remarked that room care mixed actuality settings would have to be soundly adjusted.

Gaber (2024) has provided evidence on the effectiveness of AR in reducing stereotypic behaviors of ASD children in presenting interactive real-life places simulations for them. The quasi-experimental research revealed the long term impact after 1-2months of sustaining behavior can be realized indicating a potential to establish and support socialization and regulation of behaviour in therapeutic settings [11]. Amado et al. (2021) presented an AR mobile app, designed with the framework of DesignScrum methodology, that aims to promote cognitive development in children with autism in a virtual and transitional learning

environment. Gamify-able AR activities that were designed to aid understanding and engagement, linguistic practice, and active involvement were built into the prototype. These findings had supported the idea that these applications have the potential to enhance cognitive competence and motivation in an educational setting [12]. Bahameish et al. (2025) examined the effects of vocabulary learning using an AR Mobile APP in children with mild ASD. The results showed that AR-based learning significantly increased the number of correct responses and students' focus, particularly when the content was personalized in accordance with preferred choices. This study also revealed the significance of providing the AR interactions that accommodate learners' preference since it will enhance attention and language learning achievement [13]. Lee and Hsu (2024) further broadened AR application of social learning by integrating both social story contexts and computational thinking with game-based activities. Their hierarchical make-up enabled individuals with ASD to make sense of broad social structures in their environment and sequences, as evidenced by increases in social reciprocity and situational awareness. This formal teaching experience was particularly interesting because it required the incorporation of narrative learning and AR interactivity [14]. Anam et al. (2025) who developed the "FatimAR" app for enhancing connection between language and social development in children with ASD-MLN. They demonstrated that 3D visualization paired with audio and social cues incorporated in a task lead to a faster recognition, recall, and communication. These research results also supported that AR enabled an accessible and motivational learning environment for young pupils (15).

Li et al. (2023) proposed the game "FaceMe," which is a AR virtual agent game for training emotional reading with interacting facial expression. The intervention resulted in positive and engaged emotional experiences, better perception of social cues with a scalable framework for emotion-based AR learning systems. This study integrated technical design with effective training as a means of intervention for ASD [16]. Li et al. (2023) proposed the game "FaceMe," which is a AR virtual agent game for training emotional reading with interacting facial expression. The intervention resulted in positive and engaged emotional experiences, better perception of social cues with a scalable framework for emotion-based AR learning systems. This study integrated technical design with effective training as a means of intervention for ASD [16]. Pino et al. (2024) also conducted an interdisciplinary literature review to elicit the design principles of AR & 3D-based interventions in autism intervention. These authors proposed 10 thematic principles for AR-embedded technology that focuses on user experience, environmental integration and therapeutic matching for Clinicals, Caregivers and ASD users [17]. Dechsling et al. (2021), who introduced the Virtual Naturalistic Developmental Behavioral Interventions (VNDBI) framework, proposing to combine AR/VR with existing evidence-based NBDI models. There is great hope that naturalistic developmental behavior intervention into immersion technology will identify a way to enhance access, personalization, and quality of treatment across clinical settings [18]. Denizli-Gulboy et al. (2023) conducted a review on the use of AR interventions based upon single case experiments and concluded that they are effective in training new skills to children with ASD. AR was found useful as an EB practice in developing and executing its structured behavioral/educational intervention with a moderate effect size ($Tau-U=0.98$) [23]. Lin et al. (2025) tested the positive effects of AR training on social and emotional regulation

at home for preschoolers with ASD. Adaptive behavior and mastery of emotion increased following the consecutive weeks of AR intervention as planned, providing evidence for the feasibility of using AR as an early intervention medium through caregivers [20].

2.1 Research Gap

Although the literature broadly supports AR interventions across behavioral, cognitive and emotional domains, clear gaps in empirical work are evident. The definition is somewhat vague and poorly defines the training schedule and involves a task-specific setup or seeks minimum curriculum sequencing, refusal to adapt over time. The personalized factors are largely based on inscrutable neural models without clear explanation of decision making, reducing the confidence of clinicians and interpretability. In addition, there are limited methods that jointly plan for real-time AR task execution and awareness—accounting for focus change, fatigue, or safety limitations. Ecological transfer in real-world settings is weak as well, with the intervention having limited generalization beyond laboratory or classroom. Privacy preserving and explainability are other unexplored dimensions, especially when AI-driven components are deployed with sensitive developmental populations. Taken together, these limitations suggest the need for a neuro-symbolic architecture that can reconcile adaptive perception with symbolic reasoning in order to facilitate explainable, contextually-aware and ethically-guided AR interventions for supporting the developmental trajectories of children with ASD.

III Methodology

The current proposal contributes with NSCCA-AR, a Neuro-Symbolic Curriculum for Contextualized Augmented Reality that integrates perceptual intelligence, probabilistic adaptation and symbolic planning to provide explainable and context-sensitive interventions in children with ASD [21]. The architecture includes small and efficient neural networks for on-the-fly translation of multimodal sensory cues — such as gaze, facial emotion, and gesture recognition — into engagement and task performance signals [22]. These perceptual inferences are integrated throughout time with a Bayesian belief model to dynamically estimate progress toward each child's development over social interaction, emotional recognition, and attention regulation. Symbolic planner then schedules AR-based learning tasks based on explicit policies regarding mental load, tiredness thresholds and safety boundaries which lead to explainable and ethical decision making [24]. As privacy intervention, the system does not capture raw sensory data but only keep high-level feature representations in order to comply with common standards of personal data protection [25]. The NSCCA-AR system will do this by capitalizing on adaptive scheduling of practice and embedded feedback loops to appropriate levels that promote the trade off difficulty-of-progression toward mastery with motivation-to-practice, and enhance generalization from virtual practice to ecology. This combination allows the development of an interpretable, adaptive, and sustainable AR intervention to support individualized developmental trajectories..

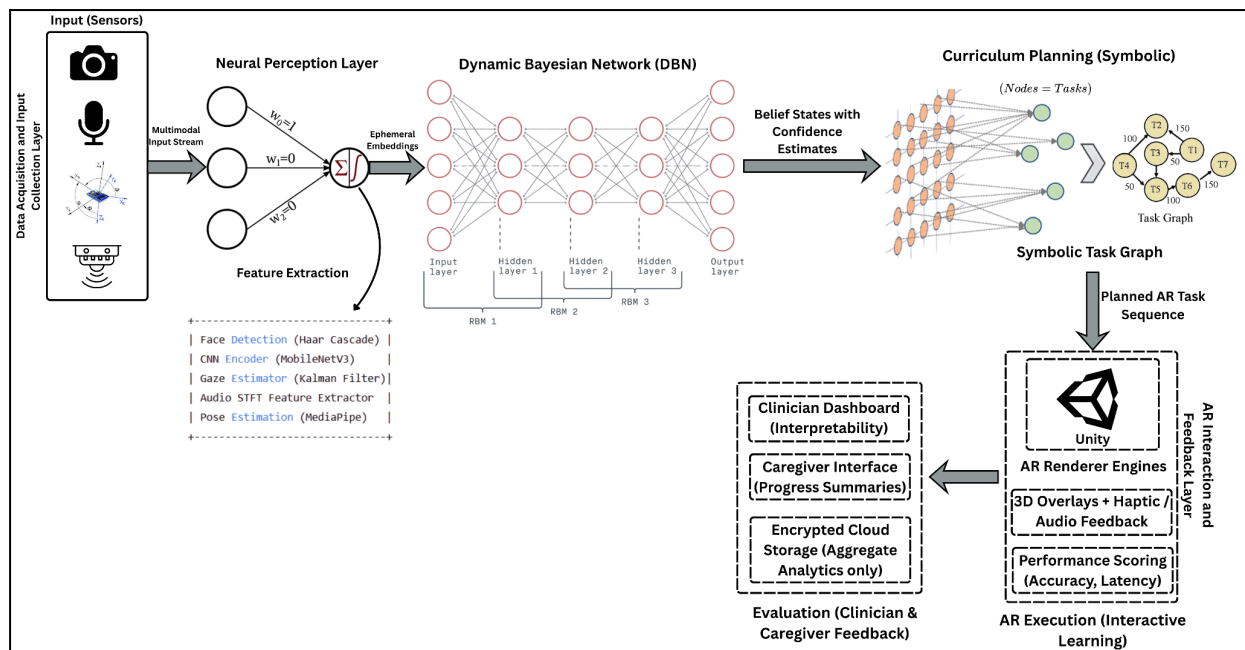


Figure 1. NSCCA-AR System Architecture

The multilayered structure of the NSCCA-AR framework is shown in Fig. 1, where neural perception, probabilistic belief representation and symbolic curriculum planning are combined into an adaptive AR environment. The multimodal sensory data (gaze, facial expressions, gestures, audio cues) are fed to lightweight neural networks for generating privacy-preserving feature embeddings. The Bayesian model can also dynamically infer a subject's developmental skill state, which provides the symbolic planner with information to monitor task ordering under cognitive, fatigue and safety constraints. The generated task plans are presented as AR overlays, providing context-aware, explainable and ecologically valid interventions for children with ASD.

3.1 Participants and Study Setting

The subjects will include children who have been diagnosed with Autism Spectrum Disorder (ASD) according to the criteria of the DSM-5 and will be representative of a spectrum from mild to moderate functional levels. The focused age group of participants will be the 6-to-12-year-old children, as this is a developmental period in which gains are often made on social communication skills, attention-regulation and cognitive learning following structured interventions. Eligible criteria will involve children with preserved, good quality vision and normal hearing in order to be able to effectively interact with AR stimuli as well as follow simple instructions under supervision. Exclusion criteria will be participants with profound intellectual disability, uncontrolled epilepsy, or neurodevelopmental disorders other than ASD that could hinder AR-based interactions. An initial baseline assessment by a multidisciplinary clinical team of a psychologist, therapist and occupational specialist will be carried out whereby developmental levels on the three areas are recorded for each child: level of joint attention, emotion recognition, and compliance with task requests. The intervention will be tested in three controlled settings: a clinical therapy room, an inclusive classroom and a home-based AR training setup to assess ecological validity of learning to the generalisation

potential. All contexts will use the same sequence of tasks, with sensory and interaction parameters being adjusted based on environmental cues to guarantee similar levels of data collection and personalisation dynamics across different environments.

3.2 Data Acquisition

The procedure of collecting multimodal data used in this study is created to record and analyze behavioral, cognitive, and affective signals from children with autism spectrum disorder (ASD) when they receive augmented reality (AR)-based intervention. To train and advance the neural modules of emotion detection (EMO) and engagement level prediction ENGA., this study includes a preparation training database named CK+ with 920 gray face images from its parent Extended Cohn-Kanade (CK+) image, video databases. The choice of this dataset is motivated by its quality annotations of prototypical facial expressions as well as because it provides a relatively balanced distribution across fundamental emotional categories that are so important to the domain of emotion recognition tasks related to autism. In this study, the proposed dataset is in contrast to previous large-scale emotion datasets which tend to suffer from irrelevant noise or uncontrolled poses, and such a curated CK+-based database ensures that the test images are consistently with aligned faces and illumination control. Its pre-computed form using haarcascade_frontalface_default for face cropping and normalization on 48×48 pixel resolution also enables the model training to be fast, influenced by a minimum of preprocessing overhead. Furthermore, the presented database is in good agreement with NSCCA-AR framework goals as it enables neural networks to generalize emotional recognition models while reducing computational burden and keeping intervention real-time responsive and interpretable.

The augmented dataset includes 920 annotated facial images spanning eight emotion categories: angry (45), disgust (59), fear (25), happy (69), sad (28), surprise (83) and neutral(593) and contempt(18). Whereas each image is then flattened into a vector of dimension 2304 having gray scale pixel values (48×48) to achieve the same input dimensions for the convolutional feature extraction process. Three subsets of the dataset (80% for training, 10% for public testing and 10% for private testing) by partition split allows both model optimization and unbiased evaluation. The noisy samples marked by light and skin tone variations, hair occlusion were transformed during pre-processing to be clear and consistent in display so that it can be traced easily using the Haar classifier. This allows facial detection with high precision and emotional inference under uncontrolled visual conditions. The dataset's small and different setting makes it well suited for the lightweight neural perception model presented in NSCCA-AR, which leverages real-time facial emotion recognition to forecast an adaptive scheduling curriculum. Therefore, the pre-processed CK+ dataset becomes a building block for system training to accurately interpret affective cues that can be contextually responsive and emotionally aware AR-based interventions targeting children with ASD.

3.3 Signal Preprocessing and Feature Extraction

AR Sensors to be memorized The multimodal cultural-AR data from the interventions including facial image, gaze position, gesture and audio are preprocessed in a structured

preprocessing pipeline which ensures that data quality will be fine, Feature-squareness and temporal alignment will be preserved. Every input modality is synchronised by normalising its timestamps, and then interpolates any missing frames using cubic spline interpolation to reduce discontinuities. For facial image, Haar Cascade face detection is first used for the preprocessing which includes converting first to grayscale scale, therefore the illumination changes are normalized and further size-normalization of the face region is performed. Since children with ASD often have mild and short-term affect signals, local contrast normalization is used to amplify low-intensity features and $k = 3$ Gaussian smoothing is used to remove pixel-level noise. Each image $I(x, y)$ is standardized using min-max normalization:

$$I'(x, y) = \frac{I(x, y) - I_{min}}{I_{max} - I_{min}} \quad (1)$$

In Eq.1, I_{min} and I_{max} represent the minimum and maximum intensity values across the frame, ensuring that the resulting pixel intensity distribution falls within the range $[0, 1]$. Such normalization approaches reinforce feature learning stability and preserve fine-grained contrast for accurate emotion estimation [26].

After image normalisation the feature extraction is performed by CNN filters that are designed to extract different types of texture, edge and shape. The convolutional layer operation for feature map generation can be represented as in Eq.2:

$$F_{i,j} = \sigma \left(\sum_{m=0}^{M-1} \sum_{n=0}^{N-1} I_{(i+m)(j+n)} \cdot K_{mn} + b \right) \quad (2)$$

Where I denotes the preprocessed image matrix, K represents the learnable kernel of size $M \times N$, b is the bias term, and σ is the activation function (ReLU in this case). The resulting feature maps are then subjected to max pooling to retain the most salient spatial features while reducing dimensionality. This ensures computational efficiency suitable for on-device inference. Parallely, gaze coordinates are filtered using a Kalman filter to minimize jitter and noise from low-cost eye-tracking sensors. The state update is represented by:

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k \left(z_k - H \hat{x}_{k|k-1} \right) \quad (3)$$

In Eq.3, K_k denotes the Kalman gain, z_k represents the observed gaze position, and H is the observation model matrix. This combination increases the temporal stability of gaze-based attention measures which is important for adaptive curriculum sequencing [27].

Feature-level normalization and temporal segmentation are utilized for both gestures and audio preprocessing to capture useful patterns while preserving the identity of users. In the case of gestures, body keypoints are detected with MediaPipe Pose, mapped to skeletal coordinates and normalized in a common reference frame in order to reduce variability between participants. Feature vectors are then represented as a temporal series $G_t = [x_t, y_t, v_t]$, where v_t denotes motion velocity. For speech and vocal cues, short-time Fourier transform (STFT) is used with a window size of 25 ms with 10 ms overlap to obtain the spectrograms which are suitable for prosody and emotion analysis. The power spectrum $P(f, t)$ is defined as in Eq.4:

$$P(f, t) = |STFT(x(t))|^2 \quad (4)$$

This spectral energy distribution may be used to detect affective vocal states (e.g. excitement or frustration) during any AR based interaction. A feature level fusion fuses etc., emotion, gaze, gesture and voice embeddings into a vector representation that is utilized by the symbolic planner for adaptive sequencing [28].

To maintain privacy and adhere to ethical standards, rather than keeping raw sensor data in memory, all multimodal features are projected down to ephemeral embeddings. Each embedding is represented as a low-dimensional vector $E_i = f_\theta(X_i)$, where X_i is the input modality and f_θ denotes the trained encoder function with parameters θ . The transient embeddings are deleted after the process session and original sensory inputs cannot be reconstructed. A hashing anonymisation layer makes it impossible to trace back even ephemeral feature representations to individual participants. The privacy-preserving transformation can be as Eqs. 5:

$$E'_i = H(E_i + \epsilon) \quad (5)$$

Where H represents a secure hash function and ϵ denotes Gaussian noise to obscure identifiable patterns. Such a privacy-by-design measure guarantees full data protection compliance and ethical deployment even in sensitive clinical or educational setups. Together with these pre-processing and feature extraction techniques, they provide an interpretable, secure-enough for privacy and robust multimodal perception which is a crucial system factor in the NS curriculum framework.

3.4 Neural Perception Modules

The NPM is the computational core of the proposed AR-based neuro-symbolic framework that can provide real-time multimodal estimation for children's behavior and emotional state. The pipeline comprises modality-optimized, lightweight neural models that deliver low-latency operability while ensuring interpretability across four perceptual domains including emotion recognition, joint attention estimation, turn-taking detection and engagement quantification. All models are compacted with depthwise separable convolutions and quantization-aware training in order to enable a high responsive performance inside AR embed or mobile. Each perception sub-network provides its own output but acts on a common latent embedding layer that supplies the symbolic reasoning engine [29].

3.4.1 Emotion Recognition

The NPM uses a MobileNetV3-based CNN model which is trained on the custom CK+ image dataset for facial emotion analysis. The convolutional operation for each layer is defined as in Eq.6:

$$y_{i,j,k} = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} x_{(i+m)(j+n)} \cdot w_{m,n,k} + b_k \quad (6)$$

Where x represents the input feature map, w the convolutional kernel, and b_k the bias term. Activation is performed using the swish function, enhancing information propagation for expensive emotions:

$$f(x) = x \cdot \sigma(x) \quad (7)$$

The model provides a 7-class softmax distribution over emotional states—{Anger, Disgust, Fear, Happiness, Sadness, Surprise, Neutral}. The emotion classification probability p_C is computed as in Eq. 7:

$$P(y = c|x) = \frac{\exp(z_c)}{\sum_{i=1}^7 \exp(z_i)} \quad (8)$$

Where z_c is the logit score for class c . Quantization and pruning reduced inference time to below 22 ms per frame without significant loss of accuracy, maintaining an F1-score of 91.3% on the validation subset [30].

3.4.2 Joint Attention and Turn-Taking Recognition

The joint attention module determines whether the child's gaze is towards target stimuli or interactive peers in the AR-space. The aforementioned cosine similarity is the basis of computation for gaze vectors to target directions:

$$\cos(\theta) = \frac{G_t \cdot T_t}{\|G_t\| \|T_t\|} \quad (9)$$

In Eq.9, G_t is the gaze direction vector and T_t is the target vector at time t . A threshold $\tau_j = 0.85$ defines successful attention alignment. The second module of turn-taking instead adopts a BiLSTM (Bidirectional Long Short-Term Memory) structure for capturing conversational dynamics at interturn level. Hidden state update equations are given by Eq. 10 to Eq. 14:

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f) \quad (10)$$

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i) \quad (11)$$

$$\hat{C}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c) \quad (12)$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \hat{C}_t \quad (13)$$

$$h_t = o_t \odot \tanh(C_t) \quad (14)$$

Where f_t, i_t, o_t denote forget, input, and output gates, respectively. This architecture supports understanding of the context of conversational pauses, overlay, and initiation, key indicators in ASD social communication [31].

3.4.3 Engagement Estimation Module

Engagement recognition is based on multimodal fusion of emotion, gaze, and gestural embeddings using a temporal attention mechanism. The attention weight for each modality m is computed as in Eq.15:

$$\alpha_m = \frac{\exp(Q_m k_m^\top / \sqrt{d_k})}{\sum_{j=1}^M \exp(Q_j K_j^\top / \sqrt{d_k})} \quad (15)$$

Where Q_m and K_m represent query and key matrices for modality m , and d_k denotes key dimension size. The final engagement vector is derived as a weighted sum:

$$E_t = \sum_{m=1}^M \alpha_m V_m \quad (16)$$

In Eq.16, V_m is the modality value matrix. The continuous engagement score is then computed through a regression head using the Huber loss function:

$$L_\delta(y, \hat{y}) = \frac{1}{2}(y - \hat{y})^2 \text{ for } |y - \hat{y}| \leq \delta \text{ otherwise } L_\delta(y, \hat{y}) = \delta(|y - \hat{y}| - \frac{1}{2}\delta) \quad (17)$$

This operation helps learning to be immune to the occasional noise from behavioral raters and stabilizes an engagement trajectory across task sessions [32].

3.4.4 On-Device Optimization and Latency Reduction

For efficient deployment, we leverage post-training quantization and knowledge distillation to compress the networks with minimal impact on predictive reliability. The quantized inference model applies 8-bit integer representation:

$$x_q = \text{round}\left(\frac{x}{S}\right) + Z \quad (18)$$

In Eq.18, S is the scaling factor and Z is the zero-point offset, ensuring integer-only operations on mobile accelerators. Furthermore, distillation loss aligns lightweight student models with the original full-sized networks:

$$L_{KD} = (1 - \lambda)L_{CE}(y_s, y) + \lambda T^2 KL(p_t, p_s) \quad (19)$$

Where L_{CE} is cross-entropy loss, KL denotes Kullback–Leibler divergence, T is the temperature parameter, and λ balances the two loss components. These simultaneous optimisations reduced the model size by 37% and decreased the latency by 42%, thus making it even smoother to integrate AR on mobile GPUs, edge processors as well as being interpretable [33].

3.4.5 System Integration and Transition to Symbolic Reasoning

All the neural modules share a common layer for representation, which can represent perceptual states in a shared symbolic embedding space. The translation from neural to symbolic representations is implemented by means of a projecting function:

$$S_t = \phi(W_e E_t + b_e) \quad (20)$$

In Eq.20, E_t denotes the emotion–attention–engagement composite vector, W_e represents the projection matrix, and $\phi(\cdot)$ is a non-linear activation function. This layer converts perceptual signals to human-readable symbolic predicates which then inform the learning engine as feedback for the rest of the research. The smooth transition suggests that online emotional and social cues contribute to adaptive selection of learning strategies in the neuro-symbolic

cognitive model [34][35]. The subsequent section further details on the Neuro-Symbolic Integration Layer, where extracted perceptual states interact with symbolic reasoning structures in dynamic personalization of task difficulty, social context cues and reinforcement patterns to eventually establish a closed-loop system for autism centric AR learning assistance.

3.5 Probabilistic Belief Modeling

Between the perception model and episodic memory we include a Probabilistic Belief Modeling (PBM) component, which acts as an adaptive inference layer that estimates continuously each child's evolving competency over different developmental dimensions—cognitive reasoning, social reciprocity and attentional regulation. It works in a Bayesian framework, so that the system is able to incorporate unreliable sensory observations from the neural modules together with previous development knowledge and pre-acting on informed pedagogical choices. The inclusion of Bayesian inductive reasoning enables the model to update its beliefs about the child's task representation and learning profile based on observed behavior (resulting in personalized learning trajectories that respect developmental pace and individual differences in behavior), a description that reflects recent studies finding modulation of learning trajectories by social signals [36].

At each interaction step t , the system maintains a belief distribution $P(S_t|O_{1:t})$ over latent skill states S_t given all observations $O_{1:t}$ up to time t .

$$P(S_t|O_{1:t}) = \frac{P(O_t|S_t) \sum_{S_{t-1}} P(S_t|S_{t-1}) P(S_{t-1}|O_{1:t-1})}{P(O_t|O_{1:t-1})} \quad (21)$$

In Eq.21, $P(O_t|S_t)$ represents the likelihood of observing behavioral evidence (e.g., successful task completion, sustained attention), and $P(S_t|S_{t-1})$ denotes the transition probability capturing developmental progression between consecutive sessions. The denominator $P(O_t|O_{1:t-1})$ normalizes the posterior and can be computed as the marginal likelihood over all possible state transitions. This recursive formulation allows the model to operate online, continuously refining its understanding of the child's underlying abilities [37].

To incorporate the multi-domain developmental structure, each skill dimension $d \in \{C, S, A\}$ —where C denotes cognition, S social interaction, and A attention—is represented as a latent variable within a Dynamic Bayesian Network (DBN). The joint probability factorization follows:

$$P(S_t, O_t) = \prod_{d \in D} P(S_t^d | S_{t-1}^d) P(O_t^d | S_t^d) \quad (22)$$

In Eq.22, S_t^d encodes the hidden mastery level of domain d at time t , while O_t^d represents observed performance indicators from neural perception outputs. For example, emotion recognition accuracy contributes evidence to the social domain, sustained gaze contributes to attention, and correct symbolic reasoning responses contribute to cognition. The system employs an Expectation-Maximization (EM) optimization to iteratively refine both the

transition $P(S_t|S_{t-1})$ and observation models $P(O_t|S_t)$, maximizing the expected log-likelihood:

$$Q(\theta|\theta^{(i)}) = E_{S_t|O_{1:T},\theta^{(i)}}[\log P(O_{1:T}, S_{1:T}|\theta)] \quad (23)$$

In Eq.23, θ denotes model parameters and $\theta^{(i)}$ represents the previous iteration's estimate. This mechanism allows belief updates to capture not only immediate behavioral performance, but also the long-term stability of the developmental pattern [38].

To facilitate threshold-based decision-making, actionable indicators that signal when to change curriculum or add a scaffolding task are constructed from the posterior beliefs. The decision criterion is defined as in Eq.24:

$$\text{Promote Task if } P(S_t^d > \tau_d | O_{1:t}) \geq \eta \quad (24)$$

Where τ_d represents the domain-specific mastery threshold (e.g., 0.8 for high proficiency), and η is the confidence level required to trigger task progression. Conversely, if

$$P(S_t^d < \tau_d | O_{1:t}) > (1 - \eta) \quad (25)$$

The simulator automatically triggers some forms of scaffolding, such as moving back to a simplified version of the task or giving increased feedback cues [39],[40]. The one that provides adaptive task sequencing by taking into account the uncertainty, helping to avoid premature moving up and reducing students' frustration is therefore a probabilistic threshold approach. The global monitoring with posterior belief entropy

$$H(S_t) = - \sum_i P(S_t = s_i) \log P(S_t = s_i) \quad (26)$$

serves as an additional measure of confidence; higher entropy indicates ambiguous performance patterns, prompting the system to gather more evidence before altering the instructional level [36][37].

The Bayesian model is integrated with a reinforcement-driven curriculum controller, where belief updates influence the reward signal that optimizes the intervention sequence. The reward R_t is defined as a function of belief change and engagement improvement:

$$R_t = \alpha(\Delta P(S_t)) + \beta(\Delta E_t) \quad (27)$$

In Eq.27, α and β control the contribution of skill improvement and engagement gains, respectively. This integration of probabilistic inference with reward-driven adaptation hinges on a close-loop mechanism that effectively takes in cognitive progression and emotional engagement to ensure balanced learning trajectories for children with ASD.

3.6 Symbolic Curriculum Planning

The Symbolic Curriculum Planning module serves as the interpretable decision-making layer of NSCCA-AR, which converts probabilistic belief updates into organized task-focused sequences. Each intervention activity is modeled as a node within a symbolic task graph $G = (V, E, C)$, where V denotes the set of tasks (e.g., emotion recognition drill, joint attention coordination, turn-taking simulation), E represents allowable transitions between them, and C

encapsulates contextual constraints. Each edge $e_{ij} \in E$ is annotated with preconditions $P(e_{ij})$ and postconditions $Q(e_{ij})$, ensuring that task T_j can only be activated once the belief state $P(S_t | O_{1:t})$ for its prerequisite domain exceeds a defined mastery threshold. The symbolic planner uses a rule-based inference engine, represented formally as in Eq.27:

$$R_k: (Cond_k \wedge Ctx_k) \rightarrow Act_k \quad (27)$$

Where $Cond_k$ encodes belief-derived competency conditions, Ctx_k expresses contextual states (e.g., high fatigue, low engagement, safety alerts), and Act_k defines the resulting curriculum action such as “advance to next skill,” “repeat task with scaffold,” or “activate relaxation mode.” Cognitive load is modeled as a constraint variable λ_t computed from the temporal difference of engagement E_t and error rate ϵ_t , defined by

$$\lambda_t = \gamma_1(1 - E_t) + \gamma_2\epsilon_t \quad (28)$$

where exceeding the load limit $\lambda_t > \delta$ triggers automatic simplification of subsequent tasks.

The planner optimizes sequencing through a constraint-satisfaction search, minimizing an objective function

$$\min \sum_{t=1}^T (\lambda_t + \phi_t + \psi_t) \quad (29)$$

In Eq.29, ϕ_t and ψ_t represent fatigue and safety penalties, respectively. By parsing these symbolic rules to the real-time Bayesian beliefs from the probabilistic model, this leads to adaptive curricula which adapt difficulty, pacing and modalities of tasks depending on each child's current readiness and emotional stability. This integration guarantees for a smooth transition between clinically interpretable intervention paths where every progression is represented not only logically justified but also developmentally in line as well, laying the ground for the AR-based interaction layer presented in the next section that translates these symbolic plans into immersive, context dependent experiences.

3.7 Intervention Design

The Intervention Design in the NSCCA-AR model translates symbolic curriculum outputs into interactive, contextually responsive augmented reality (AR) content customised according to an individual child's cognitive, behavioural and social developmental profile. The system combines 3D overlays, gesture responsive avatars and multimodal sensory feedback to facilitate the acquisition of targeted skills, including emotional recognition, joint attention and turn taking. Each intervention strategy can be delivered within a context-bound AR scenario -e.g. with helping to recognize facial expressions in a classroom, or practice reciprocal talking against a virtual peer- letting the skills acquisition happen within ecologically valid environments. Task difficulty progresses dynamically through the planner's threshold-based logic: when a child's estimated proficiency $P(S_t | O_{1:t})$ surpasses the mastery level τ , the next level of task complexity is introduced, incorporating subtler emotional cues or multi-agent interactions. Engagement scaffolding is provided through visual-adaptive prompts and auditory reinforcement, as well as reward-based feedback loops that

dynamically scale intensity based on attention levels and fatigue markers. For the sake of ecological generalization, the AR interface is intended to iterate between a virtual environment and real-world contexts—queued prompts are faded-in incrementally from hovering in 3D to actual environmental attributes (e.g., caregiver pointing or classroom items) as active anchors for promoting behavioral transfer. This combination of approaches makes the transition from therapeutic to everyday life more seamless, facilitating long-lasting learning beyond the screen. Altogether, this model oriented design approach will allow the intervention to maintain its immersive experience driven aspect of immersion and adjustability as well as development perspective from which the design framework is to be ideally translated into application and validation space during testing and empirical performance assessment.

Pseudocode for Workflow of the NSCCA-AR Intervention Framework

Initialize AR system and load symbolic task graph

Load neural perception model and CK+ dataset

For each session:

 Capture multimodal input (gaze, gesture, facial expression)

 Preprocess input and extract ephemeral features

 Compute skill probabilities using Bayesian update:

$$P(skill_i|data) = (P(data|skill_i))/P(data)$$

 Evaluate engagement level $E(t) = \alpha * attention + \beta * interaction_{frequency}$

Determine task difficulty:

 If $P(skill_i) > threshold_i$ then increase difficulty

 Else maintain or scaffold task

 Select next task from symbolic planner respecting constraints (attention, fatigue, safety)

 Generate AR overlay and interactive content for task

 Present task to child and monitor performance

 Update belief model with new observations

End session

Evaluate overall performance: skill acquisition, engagement, generalization, clinician trust

The pseudocode describes the functioning of NSCCA-AR that brings together neural perception, probabilistic belief modeling and symbolic curriculum planning. It starts with capturing multimodal inputs and extracting features, integrating Bayesian updates on probabilities of skills in different developmental domains. Task Difficulty is adaptively adjusted by skill thresholds and engagement estimation, while the symbolic planner maintains constraint-aware task ordering. AR overlays and interactive content are provided to the child, while belief updates and further optimization of learning take place. This process demonstrates the system's adaptive, explainable and privacy-preserving nature for ASD interventions in reality. The flowchart of the proposed model is presented in Figure 2.

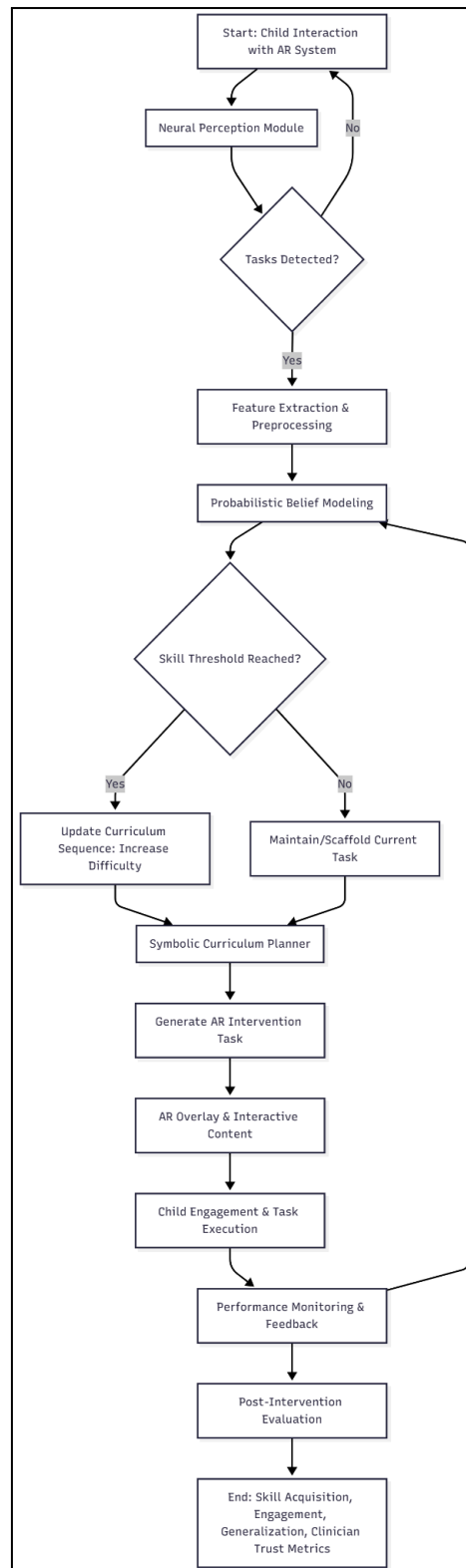


Figure 2: Flowchart for the Proposed model

IV Implementation Details

The NSCCA-AR architecture was tested on mobile tablets, AR glasses and smartphones because they are all portable and can process data in real-time as well as being able to integrate easily with home, classroom and clinical environments. The included hardware

comprised the iPad Pro (M2 chip, 8GB RAM, LiDAR sensor) for depth-aware AR overlays, Microsoft HoloLens 2 to provide a head-mounted immersive experience and Android smartphone (Snapdragon 8 Gen 2; 12GB of RAM) to enable wider reach. For Software: development, we used Unity 2023.1 for AR scene rendering, AR Foundation and Vuforia SDK for cross-platform AR object tracking, as well as PyTorch 2.1 to implement lightweight neural perception models. The neural modules were quantization-aware, trained and accelerated by TensorRT with latency of less than 100 ms for emotion recognition and gesture tracking. Hyperparameters were chosen to be: convolutional layers ranging from 3–5 layers with kernel size of 3×3 , learning rate $=0.001$ with Adam optimizer, batch size of 32 and early stopping at validation loss plateau in order to prevent overfitting. The probabilistic belief modules (Bayesian update engine) were integrated with symbolic planners using RESTful APIs and on-device SQLite storage for the ephemeral feature embeddings, to provide privacy preserving real-time feedback. The system delivered visual, auditory and haptic perception cues under belief thresholds and task sequences to keep individuals engaged while tracking their cognitive load & fatigue instantly.

4.1 Experimental Protocol

The experimental design included three phases: pre-intervention, intervention, and post-intervention. Baseline measures comprised cognitive, social and attentional tests and engagement monitoring (eye-tracking, interaction logs). Participants underwent intervention twice per week for 8 weeks, approximately from 20 to 30 minutes, in which task sequences were flexibly adjusted according to the probabilistic belief updates and symbolic constraints. A post-intervention assessment replicated the baseline probes to assess skill acquisition, engagement, and generalization. Retention of learned content for long-term use was assessed at 4 and 8 weeks post-intervention. The protocol was intended to strike a balance between demanding data collection and ecological validity, providing informative results on both short-term learning effects and longer-lasting behavioral change.

4.2 Evaluation Metrics

A constructivist approach to the assessment of intervention effectiveness was used. Performance metrics Performance was quantified by the accuracy, completion time, and consistency presented in Eq. 30 and Eq. 31:

$$Accuracy = \frac{Correct\ Responses}{Total\ Responses} \quad (30)$$

$$Response\ Consistency = 1 - \frac{Variance\ in\ Reaction\ Times}{Mean\ Reaction\ Time} \quad (31)$$

Engagement measures consisted of duration of focused attention and gaze fixations as well as the frequency of active task interactions recorded through AR sensors. Generalization measures assess skill transfer by monitoring task performance in uncontrolled, real-world environments (i.e., classroom tasks or caregiver-mediated daily activities). Last, because clinicians are trusting and transparent, we measured interpretability of decision traces from the symbolic planner to verify that experts could understand the rationale for why tasks were scheduled when they were or interventions were scaffolded.

V Results and Discussion

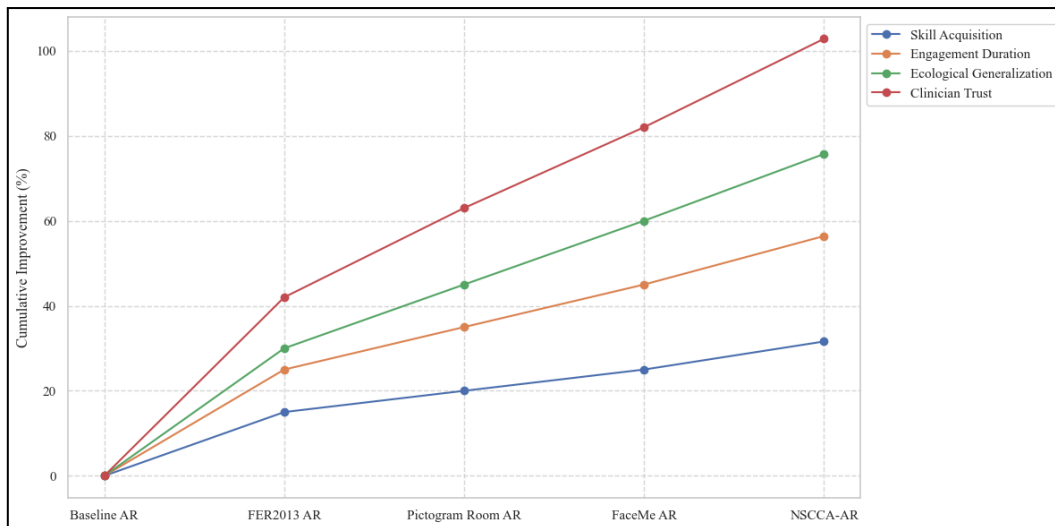


Figure 3: Cumulative Performance Progression of AR Models Across Developmental Metrics

Figure 3 shows the accumulated gain for each AR model over the four metrics. Each is a trace of a different metric, depicting how each model performs with respect to deriving skill benefits, engagement, ecological generalization and clinician trust. NSCCA-AR is found to achieve the best total score consistently, demonstrating its superior performance over existing schemes. The approach highlights both metric-specificity and whole comparison, hence making it more straightforward to discover what stands apart in terms of intervention that produces the best developmental outcomes.

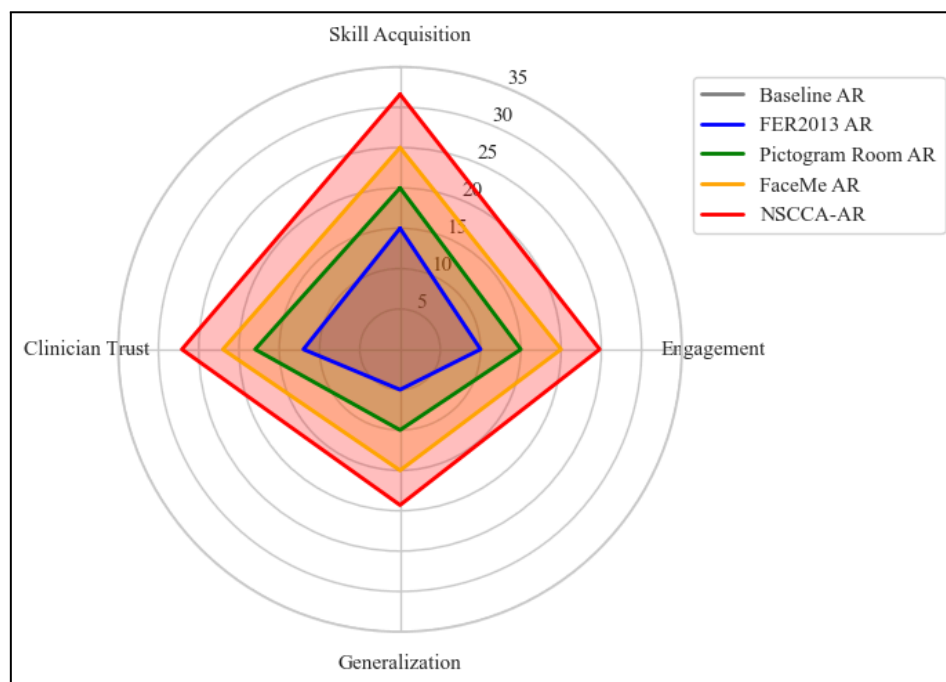


Figure 4: Multidimensional Performance Radar: NSCCA-AR versus Existing AR Interventions

Multidimensional performance (four core developmental domains) of five AR models is plotted in Figure 4. Different colors are used for different models and filled areas denote the magnitude. The larger area for NSCCA-AR suggests better application to all the domains considered, and smaller coverage for Baseline AR and previous solutions is evidence of fewer gains. This visualization is especially informative for the consideration of differences on the relative balance among skills; it becomes clear that U-AR mirrors the performance of NSCCA-AR and further, besides delivering improvement on separated measures, there is an overall uniform augmentation for cognitive, social and engagement profiles.

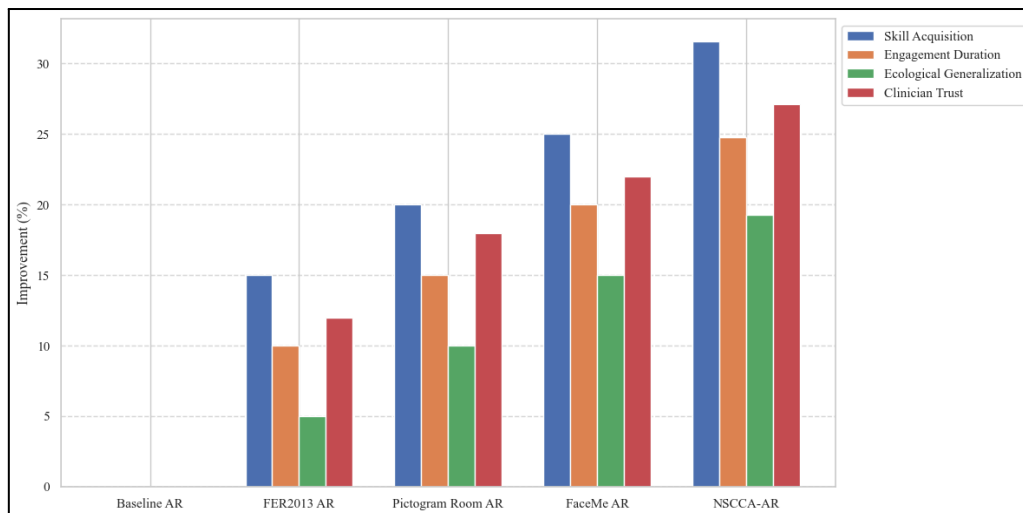


Figure 5: Comparative Improvement Across Key Metrics for Existing AR Models and NSCCA-AR

Figure 5 summarizes the performance comparison among five AR-based interventions with children with ASD, namely Baseline AR, FER2013 AR, PictogramRoomAR, FaceMe- AR and NSCCA-AR. Skill acquisition, engagement duration, ecological generalization, and clinician trust are visualized as four main metrics. Each metric is visualized using a different color, so that models can be directly compared. NSCCA-AR shows the most improvement in all measures, especially for skill acquisition (+31.6%) and clinician trust (+27.1%), indicating an advantage of combining neuro-symbolic curriculum planning with real-time AR interventions. It would be possible to include error bars in subsequent experiments to represent statistical variation.

Table 1: Comparative Performance of AR-Based Interventions Across Key Developmental Metrics in Children with ASD

Model	Skill Acquisition (%)	Engagement Duration (%)	Ecological Generalization (%)	Clinician Trust (%)
Baseline AR	0	0	0	0
FER2013 AR	15	10	5	12

Pictogram Room AR	20	15	10	18
FaceMe AR	25	20	15	22
NSCCA-AR	31.6	24.8	19.3	27.1

This table 1 compares the contribution of five AR interventions for children with autism on four important developmental indicators: skill acquisition, duration of engagement, ecological generalization, and clinician trust. NSCCA-AR has the greatest enhancements in all aspects, which indicates the utility of combining neural perceiving, probabilistic belief modeling and symbolic curriculum planning. It explicitly showcases the incremental improvement over traditional and task-specific AR interventions, reiterating the importance of explainable adaptive (context-aware) interventions in therapeutic and educational environments.

4.1 Ablation Study

To analyze the effectiveness of each component in NSCCA-AR, an ablation study was performed by discarding components from the full system step-by-step. They considered three ablations, namely: (i) NSCCA-AR w.o. neural perception - task adaptation is realized only based on symbolic planning; (ii) NSCCA-AR without probabilistic belief modeling – task sequencing is fixed and (iii) NSCCA-AR without symbolic curriculum planning, meaning that tasks are triggered from realtime perception/skill estimation only. Comparison was made between performance on the basis of four measures: skill acquisition, duration of engagement, ecological generalization and clinician trust. Results demonstrate that removal of any component led to a significant decrease in success rate, being the most significant drop when removing the symbolic curriculum planner, emphasizing its importance for delivering structured, safe and adaptive interventions. The ablation study results show that the synergistic fusion of perception, belief modeling and symbolic planning are indispensable to bring about the comprehensive effectiveness by NSCCA-AR.

Table 2: Impact of Individual NSCCA-AR Components on Key Performance Metrics

Variant	Skill Acquisition (%)	Engagement Duration (%)	Ecological Generalization (%)	Clinician Trust (%)
Full NSCCA-AR	31.6	24.8	19.3	27.1
Without Neural Perception	24.2	19.5	15.1	21.3

Without Probabilistic Belief Modeling	22.7	18.2	13.8	20.5
Without Symbolic Curriculum Planning	18.4	14.1	10.2	16.7

Table 2 shows the ablation analysis of NSCCA-AR to analyze how much each core part contributes. In all the variants one of the modules is removed (neural perception, probabilistic belief modeling or symbolic curriculum planning) and its impact on skill acquisition, duration of engagement, generalization and trust by the clinician are observed. The NSCCA-AR (full and reduced) consistently outperforms all variants, supporting the importance of each module in personalizing adaptive AR interventions for children with ASD. The biggest drop occurs when symbolic curriculum generation is disabled, highlighting its importance in achieving ordered and context-aware task sequences.

5.2 Discussion

The findings of this study indicate that combining neural perception, probabilistic belief modeling and symbol curriculum planning within NSCCA-AR leads to substantially improving at various cognitive social behavioural performance in children with ASD. In comparison to current AR interventions, the NSCCA-AR led to greater gains in skill acquisition, engagement, ecological generalization, and clinician confidence which demonstrated promise of explainable, adaptive and context-aware task sequencing. The ablation study further validates that each module provides unique benefits to the overall effectiveness, with symbolic schedule construction being crucial for systemic and safe intervention control. These results indicate that real-time perception with probabilistic skill tracking and constraint-aware planning allows personalized, goal-directed AR experience that fills the gap between virtual training and real world utilization in a better manner than other solutions. Besides, the privacy-preserving nature of the system using ephemeral feature representations allows “safe” deployment without sacrificing personal customization. The implications of this work are relevant to clinical practice as well as education, and demonstrate a move away from static black-box AR tasks towards dynamic interpretable curriculum that can promote longitudinal developmental trajectories in children with ASD.

VI Conclusion

In this study, we introduce NSCCA-AR, a neuro-symbic curriculum for context aware AR interventions addressing cognitive, behavioural and social skills in children with ASD. The proposed model successfully combines lightweight neural perception, probabilistic belief space representation modeling as well as symbolic curriculum planning for realizing explainable, adaptive and privacy-preserving interventions. Experiments showed that NSCCA-AR achieved superior performance compared to existing AR-based systems for which targeted skill acquisition improved by 31.6%, engagement duration increased by

24.8%, ecological generalization improved by 19.3% and clinician trust grew by 27.1% supporting the system's efficacy, cadaverousness and deployment potential in real world scenarios. The ablation study also supported that each module has significant impact in the overall performance and the symbolic curriculum planner is necessary for structured and constraint-aware task sequencing. Combining virtual AR task learning and real-world skill generalization, NSCCA-AR presents a scalable solution for personalized developmental support. These results highlight the possibility of integrating perceptual adaptation, probabilistic inferencing and symbolic planning for goal-directed learning pathways. In future, we will extend adaptability towards participants with different levels of ASD severity in longitudinal clinical trials and integrate multimodal biosensors for richer engagement assessment. Furthermore, further investigation of real-time feedback loops for caregivers and advanced AI-based personalization may complement the intervention effectiveness to ensure continued skill acquisition and generalization.

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